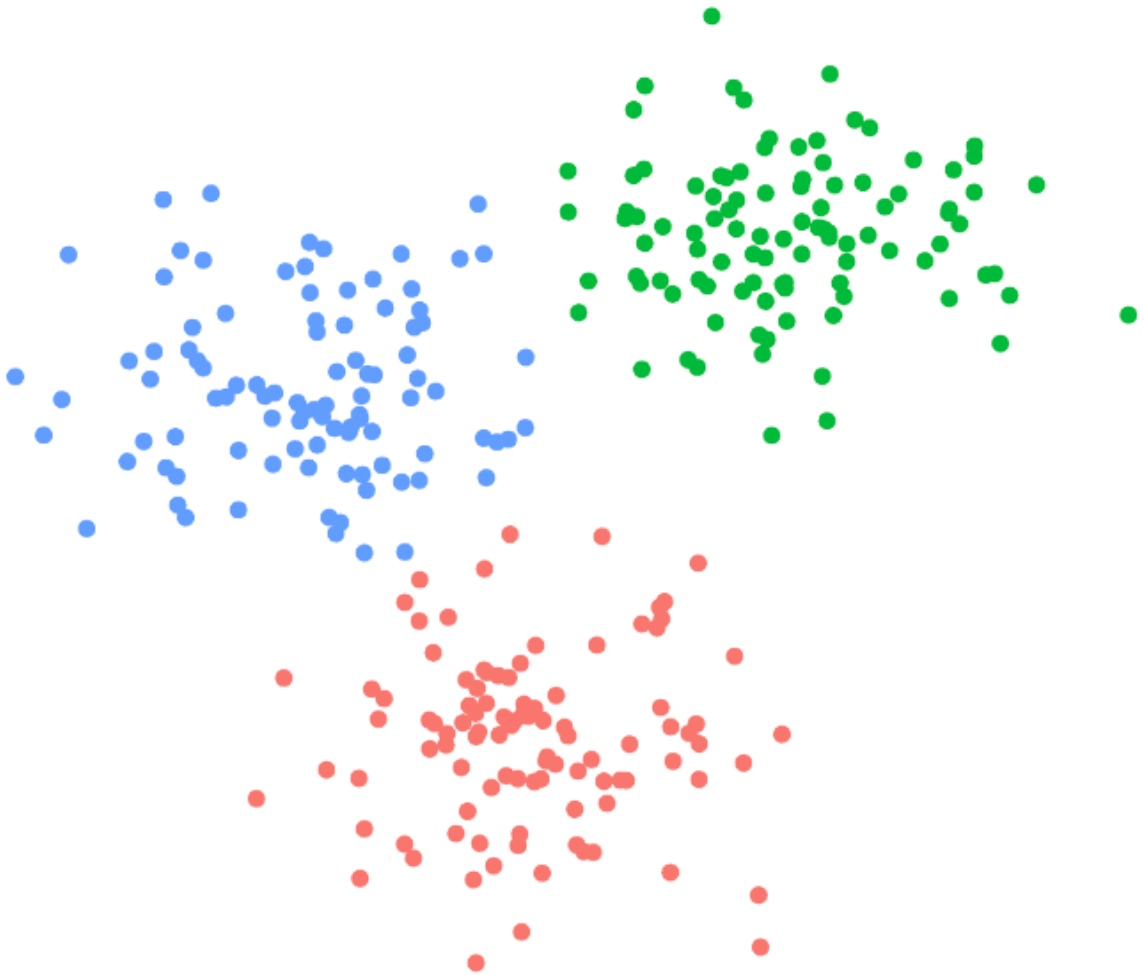
DBSCAN

Density based clustering

DBSCAN

- Density-Based Spatial Clustering of Applications with Noise

Look at this sample labeled dataset



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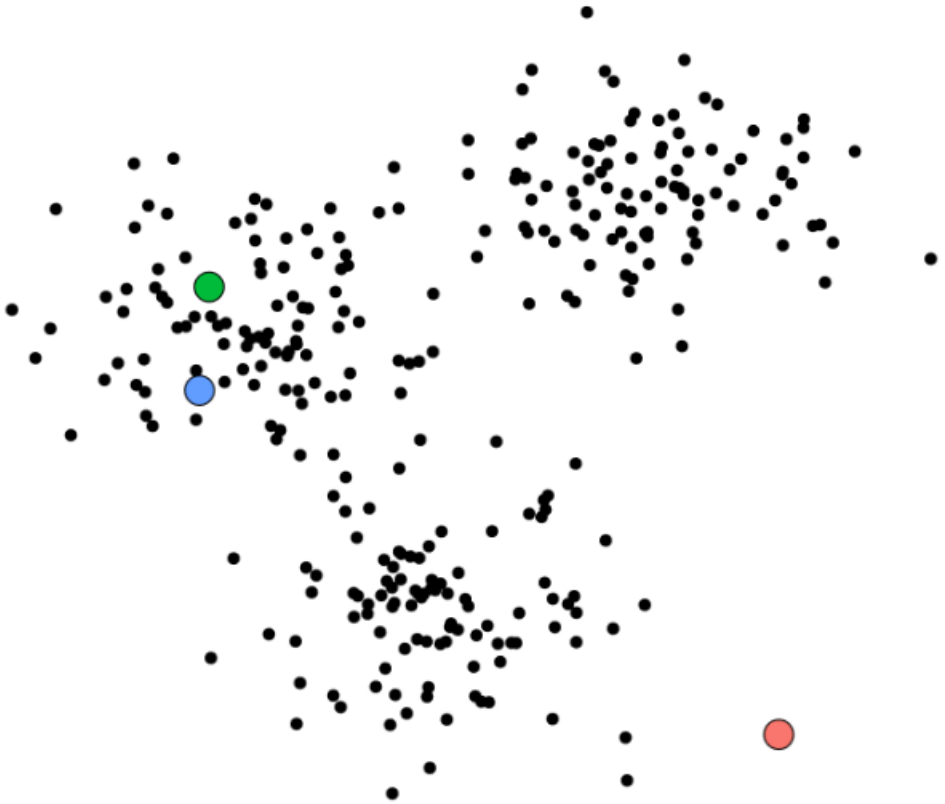
How k-means cluster this data with $k = 3$?

Start with k randomly chosen centroids



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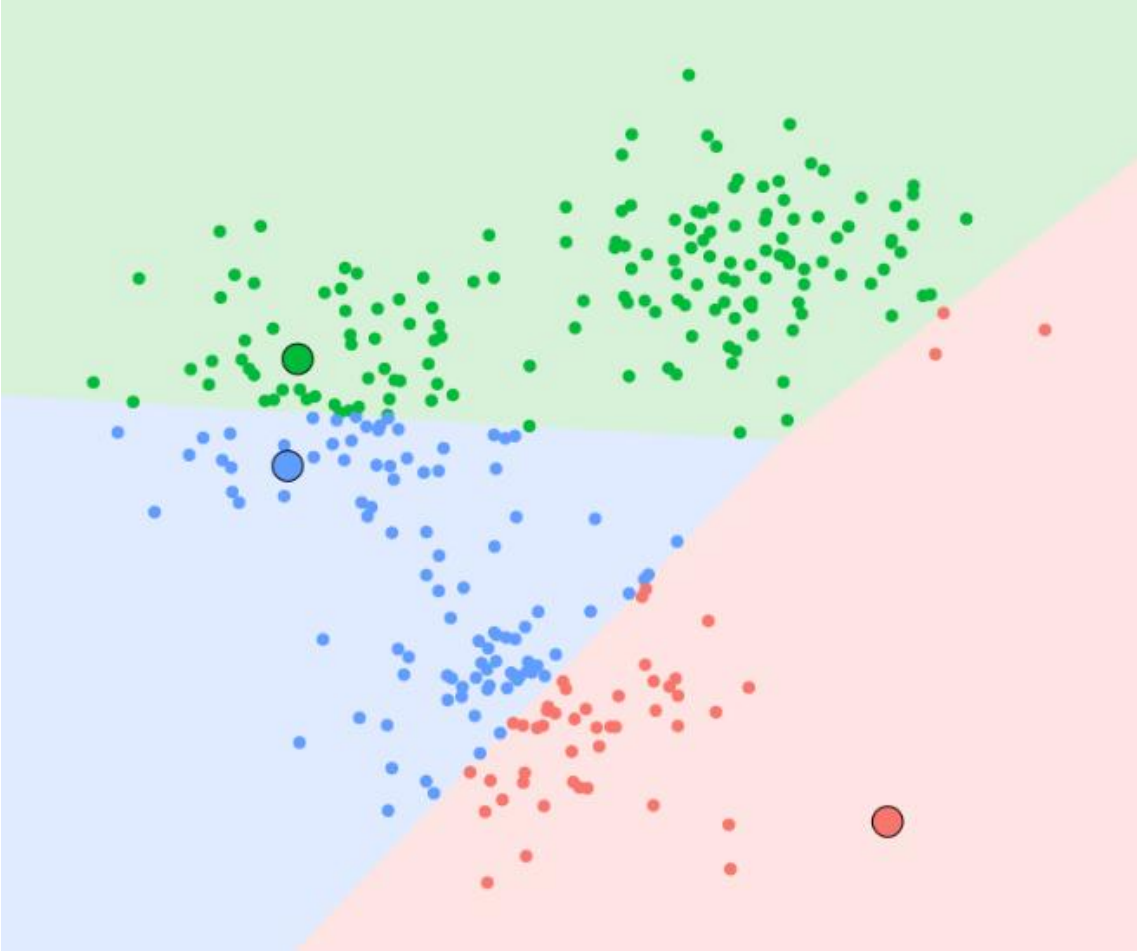
How k-means cluster this data with $k = 3$?

Assign data points to clusters by the shortest distance to any mean



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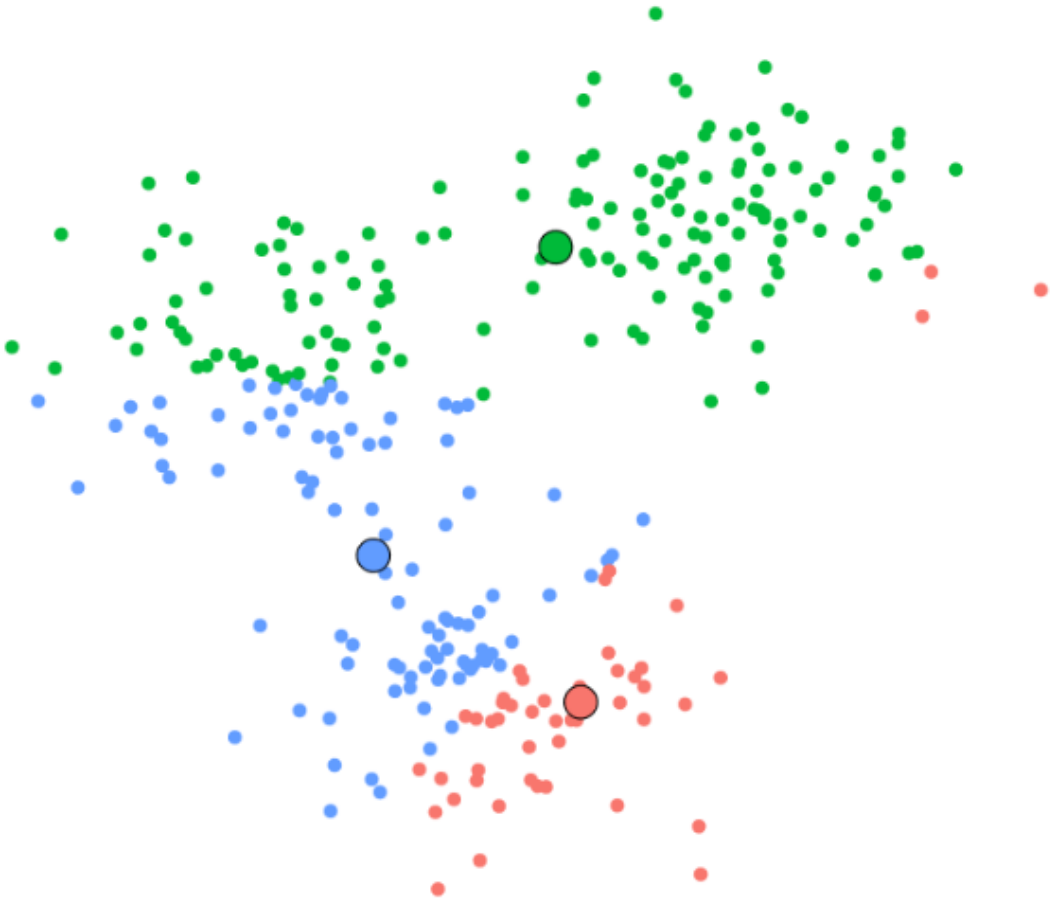
How k-means cluster this data with $k = 3$?

Update the centroids



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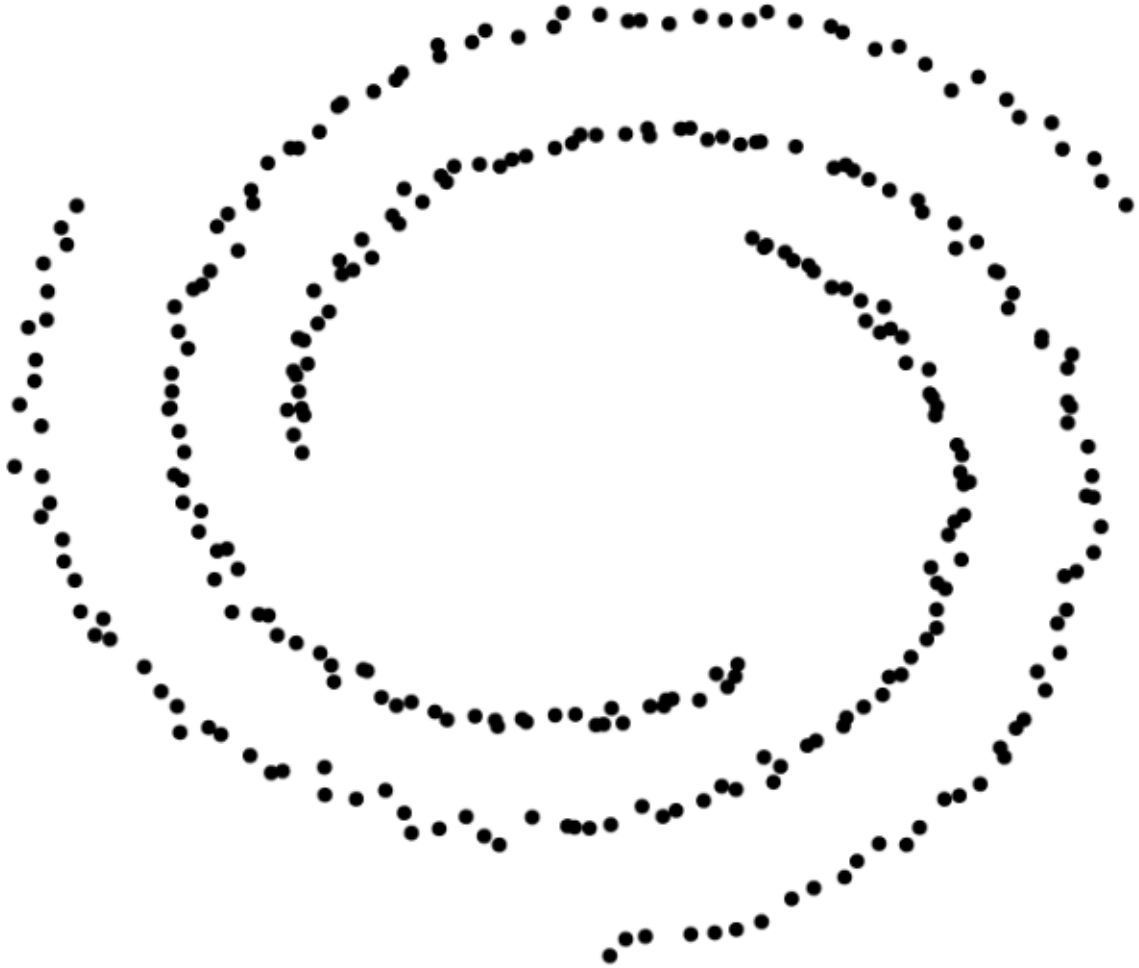
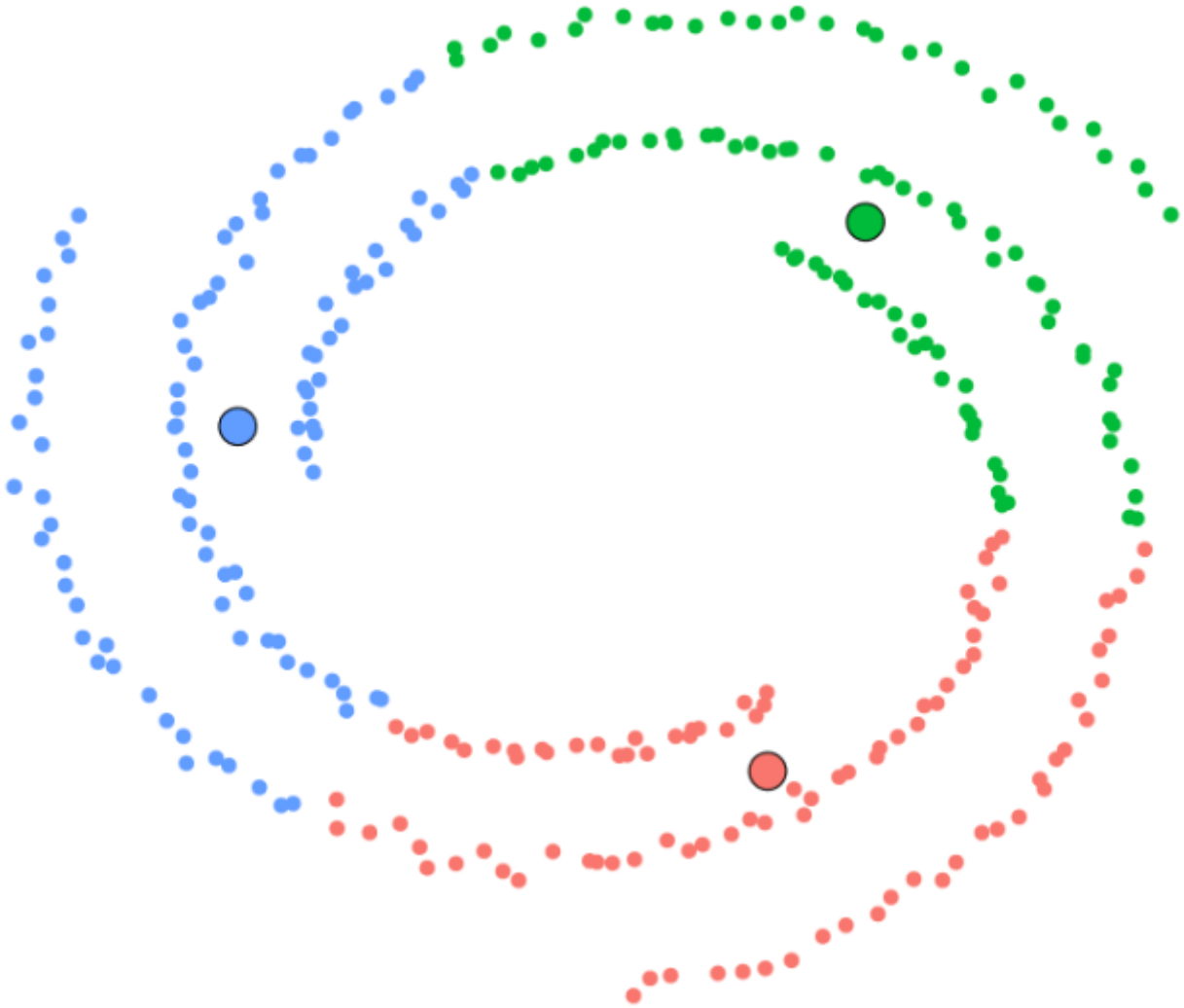
How k-means cluster this data with $k = 5$?

Repeat the steps until convergence

How about this dataset?

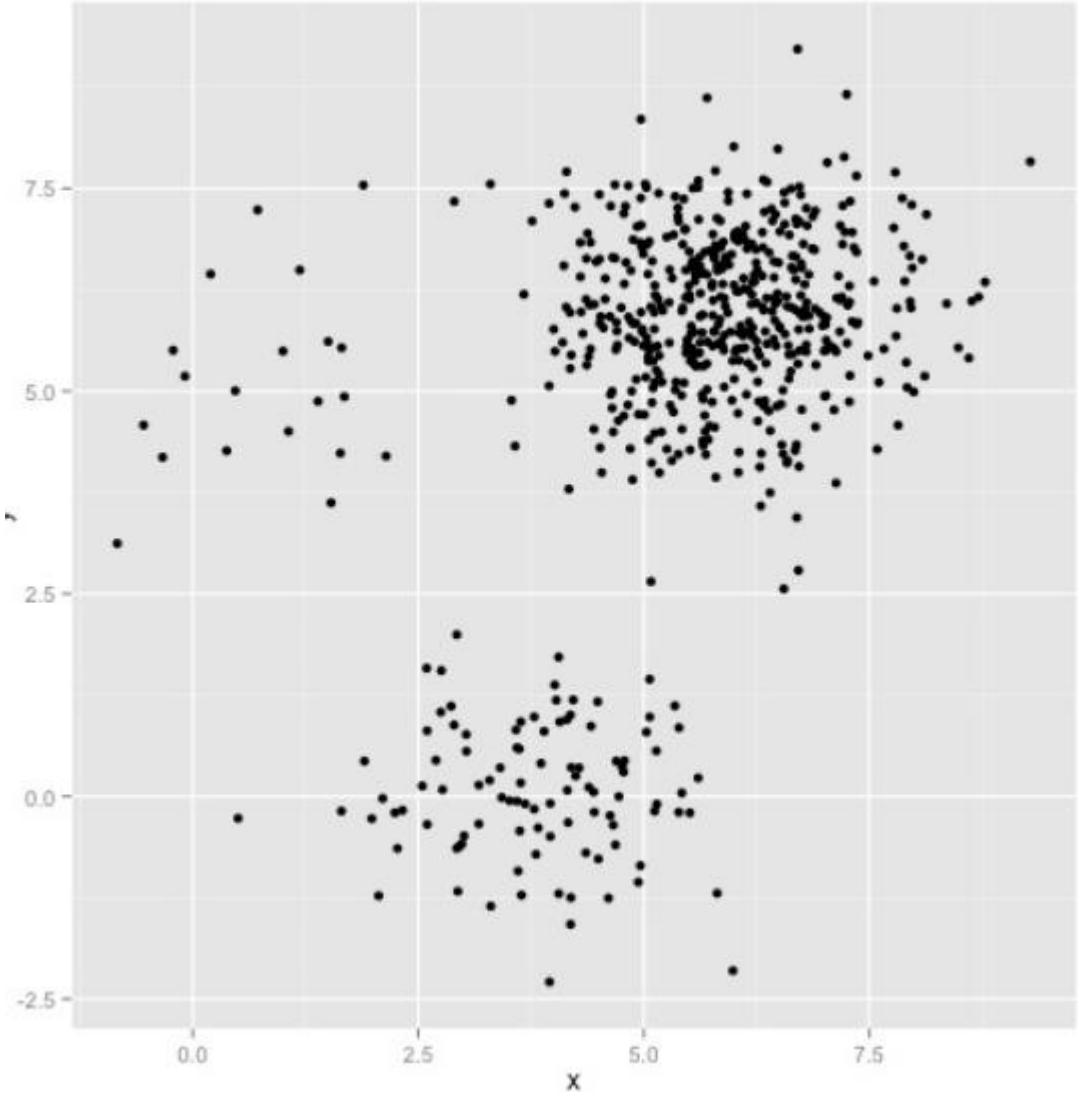


How about this dataset?



Let’s look at this example

How many clusters can you see in this plot?



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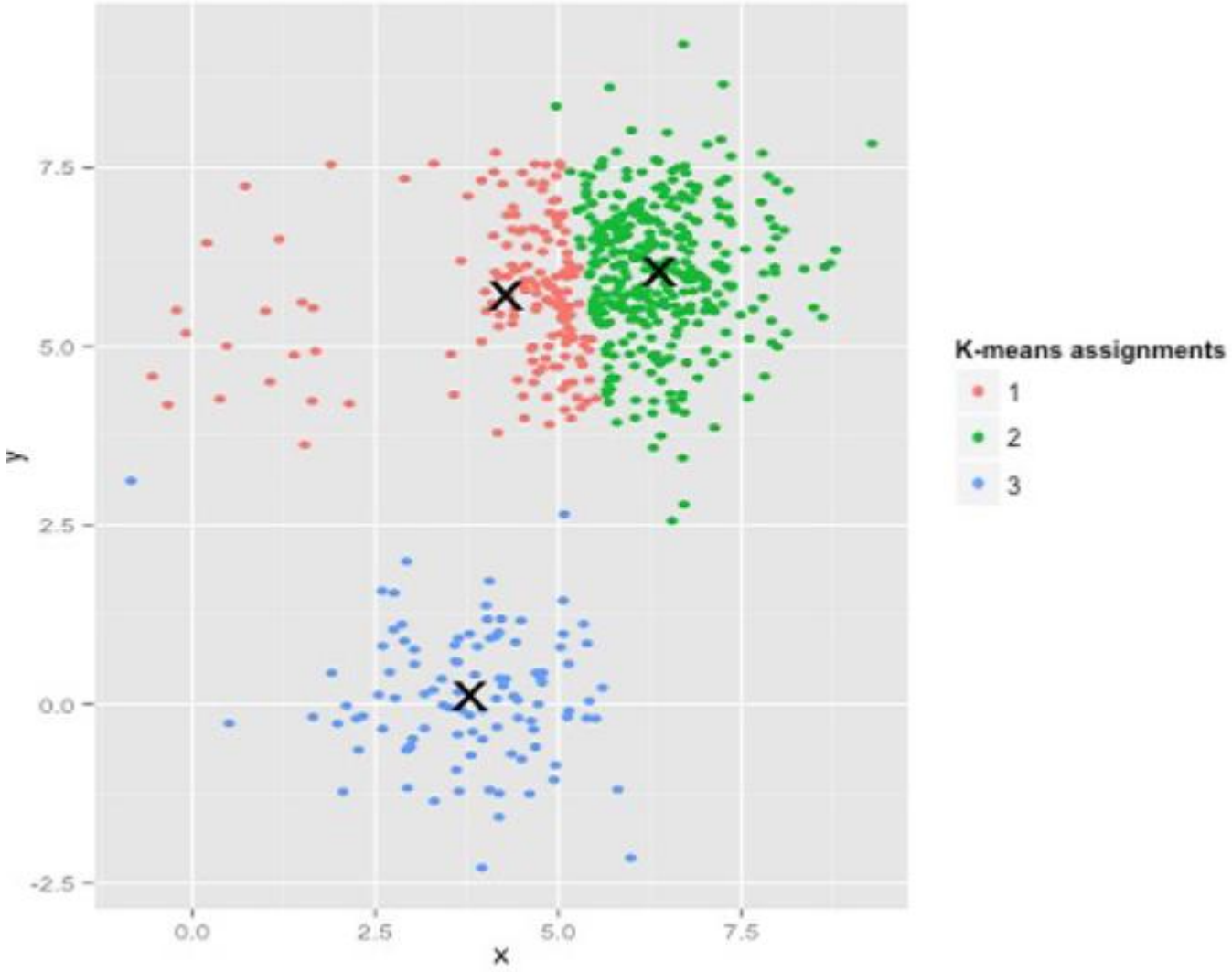
Let’s look at this example

K-means results



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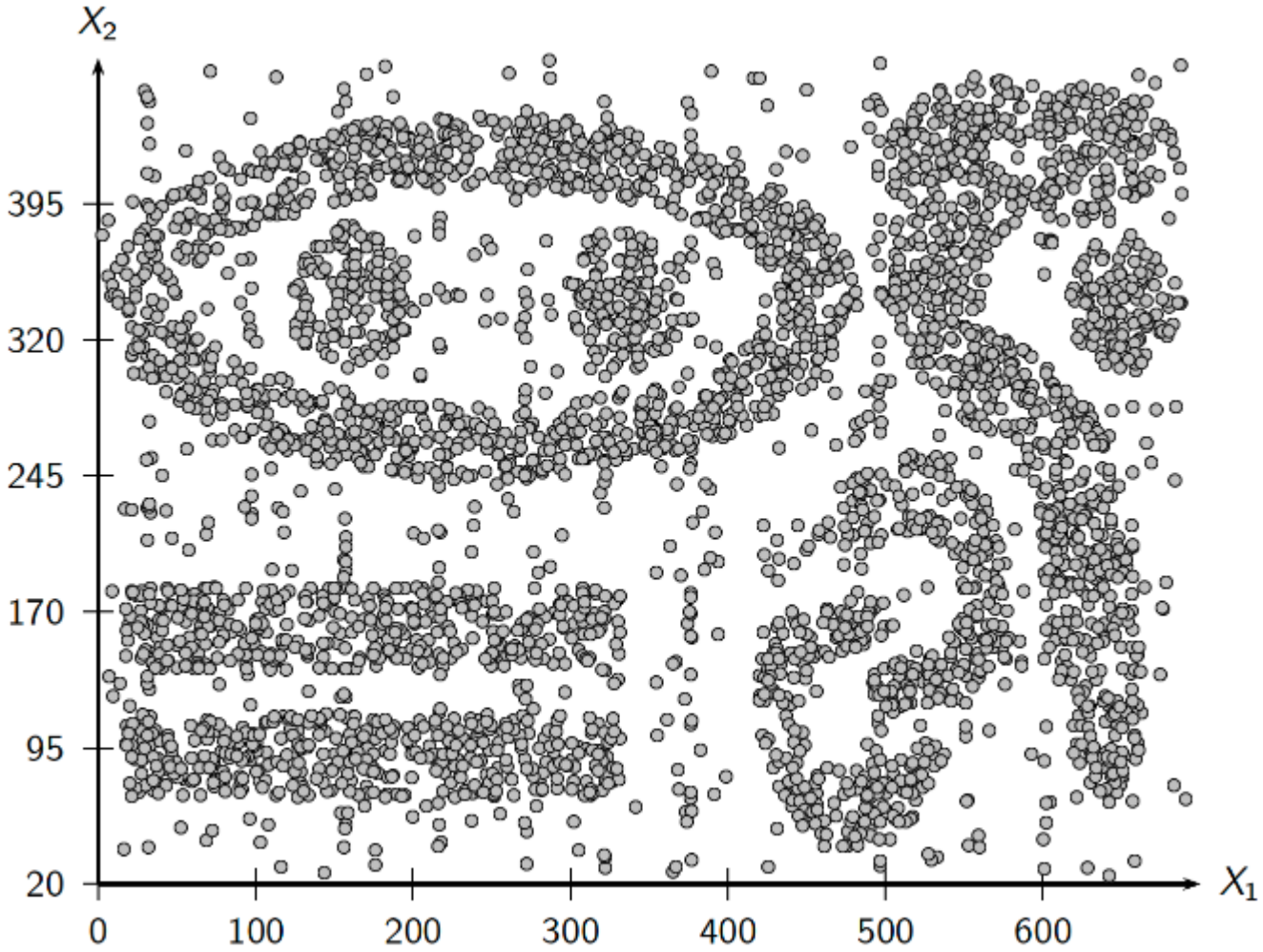
How about this synthetic dataset?



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K-means limitations

- It assumes the clusters are of convex shape.
- Sensitive to outliers.
- When clusters are non-convex, two points in two neighborhood clusters might be closer than two points in the same cluster.
- Density based methods are able to find non-convex clusters, where distance-based methods may have difficulty.
- K-means time complexity $O(tnkd)$

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K-means Demo

- https://www.naftaliharris.com/blog/visualizing-k-means-clustering/

DBSCAN approach

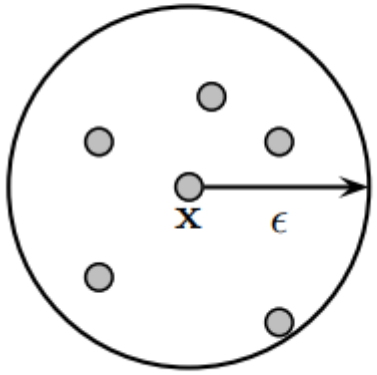
- Density-based Spatial Clustering of Applications with Noise (DBSCAN)
- Define a ball of radius ε around a point $x \in \mathbb{R}^d$, called the ϵ-neighborhood of x:

$$\boxed{N_\epsilon(x) = B_d(x, \epsilon) = \{y \mid \delta(x, y) \leq \epsilon\}}$$

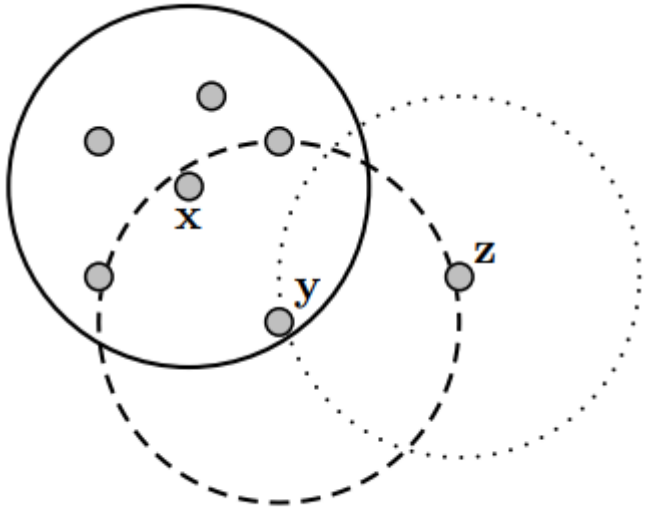
- Here $\delta(x, y)$ represents the distance between points x and y, which is usually the Euclidean distance.
  - Other distance metrics can be used as well.
- We say that x is a core point if there are at least minpts points in its ϵ-neighborhood, i.e., if $|N_\epsilon(x)| \geq minpts$.
  - minpts is a user defined local density or frequency threshold .
- A border point does not meet the minpts threshold, i.e., $|N_\epsilon(x)| < minpts$, but it belongs to the ϵ-neighborhood, or core points z, that is, $x \in N_\epsilon(z)$.
- If point is neither core nor border point, then it is called a noise point or an outlier.

Core, border and noise points

Suppose minpts = 6



(a)



(b)

(a) Neighborhood of a Point

(b) Core, Border, and Noise Point

The DBSCAN approach

- A point x is directly density reachable from another point y if $x \in N_\epsilon(y)$ and y is a core point.
- A point x is density reachable from y if there exists a chain of points $x_0, x_1, x_2, \ldots, x_L$, such that $x = x_0$ and $y = x_L$, and x_i is directly density reachable from x_{i-1} $\forall i \in [1 \ldots L]$. In other words, set of core points leading from y to x.
- Two points x and y are density connected if there exists a core point z, such that x and y are density reachable from z.
- A density-based cluster is defined as a maximal set of density connected points.

The DBSCAN approach

- DBSCAN computes the ϵ-neighborhood $N_\epsilon(x_i)$ for each point x_i in the dataset D, and checks if it is a core point. It also sets the cluster id, $id(x_i) = \emptyset$ for all points, indicating that they are not assigned to any cluster.
- Starting from each unassigned core point, the method recursively finds all density connected points, which are assigned to the same cluster.
- Some border points may be reachable from core points in more than one cluster; they may either be arbitrarily assigned to one of the clusters or to all of them (if overlapping clusters are allowed).
- Points that do not belong to any cluster are treated as outliers or noise.
- Each DBSCAN cluster is a maximal connected component over the core point graph.
- DBSCAN is sensitive to the choice of ϵ, in particular if clusters have different densities.

The DBSCAN algorithm

Algorithm 15.1: Density-based Clustering Algorithm

```
DBSCAN (D, ε, minpts):
Core ← ∅
foreach x_i ∈ D do // Find the core points
    Compute N_ε(x_i)
    id(x_i) ← ∅ // cluster id for x_i
    if N_ε(x_i) ≥ minpts then Cores ← Cores ∪ {x_i}
k ← 0 // cluster id
foreach x_i ∈ Core, such that id(x_i) = ∅ do
    k ← k + 1
    id(x_i) ← k // assign x_i to cluster id k
    DensityConnected (x_i, k)
𝒞 ← {C_i}_{i=1}^k, where C_i ← {x ∈ D | id(x) = i}
Noise ← {x ∈ D | id(x) = ∅}
Border ← D \ {Core ∪ Noise}
return 𝒞, Core, Border, Noise

DensityConnected (x, k):
foreach y ∈ N_ε(x) do
    id(y) ← k // assign y to cluster id k
    if y ∈ Core then DensityConnected (y, k)
```

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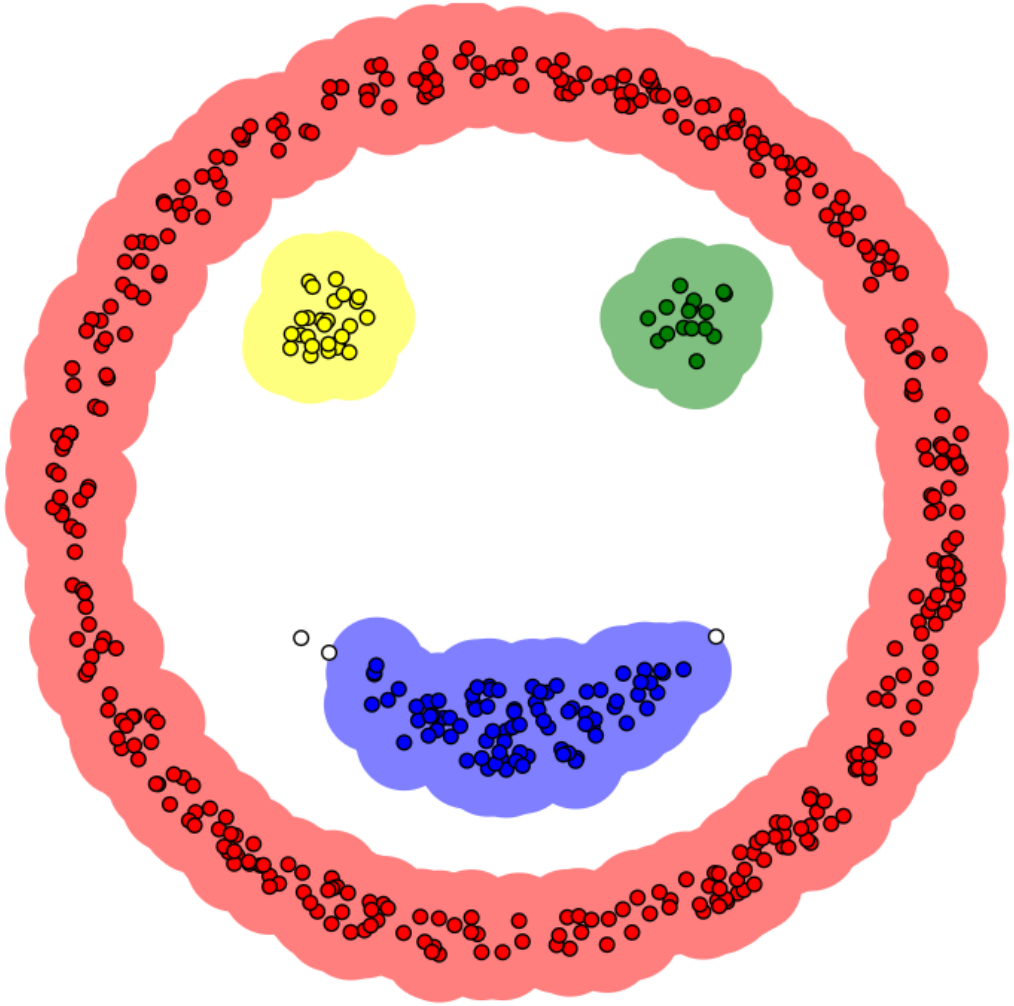
Density based clusters $\epsilon = 1.0$ and $minpts = 3$



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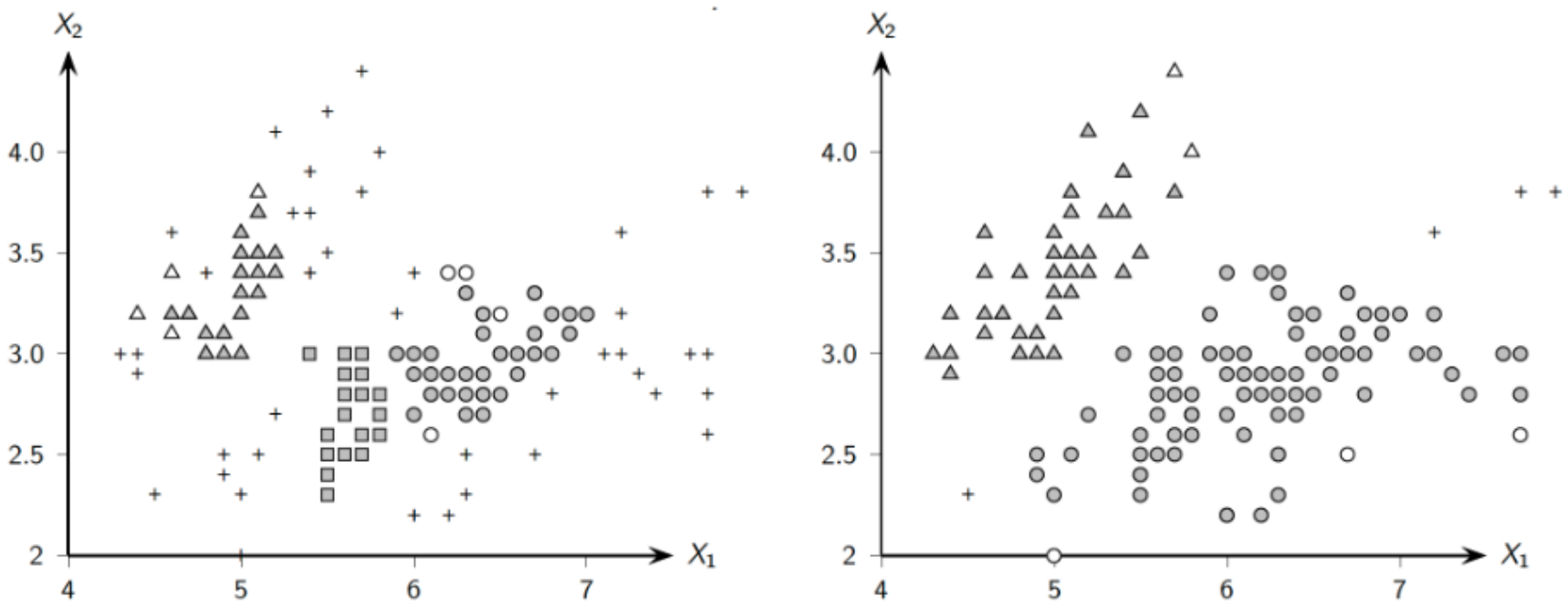
DBSCAN visualization

- https://www.naftaliharris.com/blog/visualizing-dbscan-clustering/

Disadvantages of DBSCAN

- Suffers from curse of dimensionality.
  - In high dimensions ϵ-neighborhood is meaningless
  - All the points fall closer to each other.
- Approximate appropriate values for ϵ and $minpts$ could be challenging.
- Finding clusters with different densities is difficult.

DBSCAN clustering IRIS dataset



(a) $\epsilon = 0.2,\ minpts = 5$

(b) $\epsilon = 0.36,\ minpts = 3$